Data Structures - Question Bank
Course: Data Structures | B.Tech CSE | Semester 2 | Year 1

1. Which data structure works on the FIFO principle?
a) Stack
b) Queue
c) Tree
d) Graph
Answer: b

2. Which operation removes the top element of a stack?
a) push
b) pop
c) peek
d) enqueue
Answer: b

3. What is the time complexity of binary search?
a) O(n)
b) O(log n)
c) O(n log n)
d) O(1)
Answer: b

4. Which data structure uses the LIFO principle?
a) Queue
b) Stack
c) Array
d) Hash table
Answer: b

5. A node in a singly linked list typically contains:
a) data only
b) data and a pointer
c) pointer only
d) an index
Answer: b

6. Which traversal visits left subtree, then root, then right subtree?
a) Preorder
b) Inorder
c) Postorder
d) Level order
Answer: b

7. What is the average lookup time of a hash table?
a) O(n)
b) O(1)
c) O(log n)
d) O(n squared)
Answer: b

8. Which sorting algorithm has O(n log n) worst-case time?
a) Bubble sort
b) Insertion sort
c) Merge sort
d) Selection sort
Answer: c

9. A stack can be implemented using:
a) an array only
b) a linked list only
c) both an array and a linked list
d) neither
Answer: c

10. A perfect binary tree with n levels has at most how many nodes?
a) 2n
b) 2^n - 1
c) n squared
d) 2n - 1
Answer: b

11. Explain the difference between a stack and a queue with examples.
(Write your answer in the space provided.)

12. Describe how merge sort works and state its time complexity.
(Write your answer in the space provided.)